SARA REZAEIMANESH

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PROFESSIONAL SUMMARY

Ph.D. student in Computer Science specializing in machine learning and natural language processing. My research focuses on developing approaches for overcoming data scarcity, improving out-of-distribution generalization, and advancing idiom translation and creative capabilities in large language models. I am also interested in linguistic phenomena in low-resource languages and creative text generation.

PUBLICATIONS

[1] [paper] **What LLMs Know but Don't Say: Non-Generative Prior Extraction for Generalization,** *COLM* (2026)
S. Rezaeimanesh, M. Ghassemi

[2] [paper] **Large Language Models for Persian ↔ English Idiom Translation,** *NAACL* (2025)
S. Rezaeimanesh, F. Hosseini, and Y. Yaghoobzadeh

[3] [paper] **Ordinal Gates, Cardinal Bets: Matching LLM Confidence to the Financial Decision Operator** *ICAIF* (2026)
R. Singh, **S. Rezaeimanesh**

[4] [paper] **Lost in the Long Tail: Improving Creativity and Coherence in Language Generation,** *Under Review for ICLR* (2027)
S. Rezaeimanesh, E. Liu, S. Bhat

[5] [paper] **INCONVENIENT: A Benchmark to Assess LLM Generalization Capabilities from Convenience Samples,** *Under Review for ICLR* (2027)
S. Rezaeimanesh, C. Klochko, K. Thind, M. Ghassemi

EDUCATION

MICHIGAN STATE UNIVERSITY — **East Lansing, MI**
PhD of Computer Science — *January 2025–2030*
GPA: 4.0/4.0

UNIVERSITY OF TEHRAN — **Tehran, Iran**
Bachelor's of Computer Engineering — *August 2019–May 2024*
GPA: 19.0/20 (Ranked among the top 5 students of my graduating class.)
Thesis: Large Language Models for Persian ↔ English Idiom Translation (Winner of the "Best Undergraduate Project Award")

PRESENTATIONS AND OUTREACH

- Featured in *The Detroit News* for developing an AI-augmented patient-advocate application — *2026*
  - [link] Article: MSU computer science professor teaches use of AI combined with human knowledge
- Invited panelist – "AI, Information Systems, and Knowledge" event at MSU — *2026*
- Invited speaker – Women in Computing at MSU: Incorporating AI into professional workflows — *2025*

RESEARCH EXPERIENCE

MICHIGAN STATE UNIVERSITY — **East Lansing, MI**
Research Assistant in the Human Augmentation and Artificial Intelligence Lab — *January 2025–Present*

- **Non-Generative Prior Extraction for Generalization:** *Publications [1], [2], [5]*
  - Created a comprehensive benchmark for assessing LLM generalization capabilities when the training set is a convenience sample.
  - Developed a **deterministic framework that extracts informative Bayesian priors from LLMs** to generalize patterns learned from small datasets, to larger, out-of-distribution populations resulting in the second publication.

  - **Probed LLM confidence on feature-outcome relationships through paired semantic queries** to construct priors for **tabular classification** tasks.
  - Demonstrated **improved predictive performance and posterior calibration across seven out-of-distribution** datasets in healthcare, finance, and social science domains.

- **Increasing the generation of rare multiword expressions in LLM-generated creative texts (CMU & UIUC Collaboration):** *Publication [4]*
  - Co-authoring in a research project investigating how large language models internally **process idiomatic expressions versus literal language** through mechanistic interpretability techniques
  - Devising a new look-ahead based sampling method that increases the probability of rare multi-word expression generation, and increases LLM creativity.

- **Matching LLM confidence to the financial decision operator:** *Publication [3]*
  - Co-authored and mentored the undergraduate lead author on a study showing that LLM confidence scores are **not independently deployable**: their value depends on whether the downstream decision uses their rank or their magnitude.
  - Formalized this ordinal–cardinal distinction and showed across nine open-weight LLMs (7–32B) on Nasdaq-100 news that confidence calibration and risk control must be fitted together.

- **Patient-Advocate Application:**
  - Leading a team of five graduate and undergraduate students to build a **patient-advocate connection application** to connect patients with professional advocates, improving support for undereducated, under-served communities.
  - Building an AI-powered advocate system that schedules tasks and reminders based on appointments, monitors vital signs for abnormalities, and conducts targeted health assessments to optimize patient outcomes during treatment.

UNIVERSITY OF TEHRAN **Tehran, Iran**
Research Assistant in the Natural Language Processing Lab *August 2020–May 2024*

- Proposed novel methods to improve **LLM performance in idiom translation** by combining LLMs with Neural Machine Translation models, resulting in **significant gains in both idiom translation accuracy and fluency** for sentences containing idiomatic expressions resulting in the first publication.
- Compiled the first Persian idioms dataset, PersianIdioms, containing over 2000 Persian idioms with their meanings and example usage sentences for 700 entries.
- Won the **"Best Undergraduate Project Award"** awarded by the University of Tehran among peers for the first publication.

TEACHING EXPERIENCE

MICHIGAN STATE UNIVERSITY **East Lansing, MI**
Chief Teaching Assistant *August 2025–Present*
Designed projects and graded assignments and quizzes for the **AI Agents Course**, mentoring students in developing applications that use multiple LLM-based agents to process and execute natural language requests. Guided students in analyzing how recent research in prompt engineering influences LLM performance across different benchmarks.

UNIVERSITY OF TEHRAN **Tehran, Iran**
Teaching Assistant *August 2020–May 2024*
Designed projects, provided feedback, and reviewed students' code for courses including **Natural Language Processing, Machine Learning**, Data Structures, Design of Algorithms, and Software Engineering.

SELF-EMPLOYED **Tehran, Iran**
English Teacher *May 2019–January 2026*
Pursued my passion for teaching by instructing English to students aged 5–18 in schools, and private settings.

AWARDS

- **National License Plate Recognition Contest Award:** Developed solutions to address similarities between Persian letters in license plates; achieved the highest accuracy of 98% (a 17% increase compared to existing baselines) among contestants. — 2021
- **Kharazmi National Award:** Awarded for my novel in a national student competition. — 2017
- **National Caricature Drawing Award:** First place in national contest. — 2018

RESEARCH INTERESTS

Machine Learning. Out-of-distribution Generalization. AI agents for healthcare, finance, and education. Linguistics

TECHNICAL SKILLS

Libraries and Frameworks: PyTorch (Proficient), TensorFlow (Proficient), HuggingFace (Proficient), React (Competent), Vue (Intermediate)
Languages: Python (Proficient), C++ (Proficient), C (Competent), Java (Competent), CSS (Competent), JavaScript (Competent)

EXTRACURRICULARS

GHALAMCHI INSTITUTE OF EDUCATION **Tehran, Iran**
Full-Stack Developer *May 2020–September 2021*
Designed and developed an English learning app that enables users to improve their language skills through games and competitions, as well as a charity website for the institute, allowing users to donate and track how their contributions are utilized.

LANGAUGES

Persian (Native), English (Proficient), Japanese (Elementary)